

■■ CYBERSHIELD

Software Requirements Specification

Document CS-SRS-001 — Version 1.0

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1. Introduction

Document	Software Requirements Specification
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1.1 Purpose

This document specifies the software requirements for CyberShield, a native SIEM module for ODOO 18 Community Edition. It defines functional requirements, non-functional requirements, security requirements, and system constraints.

1.2 Scope

CyberShield provides real-time endpoint monitoring, threat detection, vulnerability management, incident response and self-contained notification system fully integrated into ODOO 18. The system covers Windows and Linux endpoints via Wazuh agents and exposes a secure REST API for event ingestion.

1.3 Definitions & Acronyms

Term	Definition
SIEM	Security Information and Event Management
HMAC	Hash-based Message Authentication Code — used for API request integrity
CVE	Common Vulnerability and Exposure — standardised vulnerability identifier
MITRE ATT&CK;	Framework of adversary tactics and techniques based on real-world observations
Zero Trust	Security model requiring strict identity verification for every device and person
RBAC	Role-Based Access Control — access granted based on assigned role
TLS	Transport Layer Security — cryptographic protocol for secure communications

2. System Overview

2.1 Three-Layer Architecture

Layer	Component	Responsibility
1 — Telemetry	Wazuh Agents (Windows/Linux)	Collect security events from endpoints, forward via TLS 1.3 with mutual certificate authentication
2 — Processing	Wazuh Server + OpenSearch	Process, correlate and classify events using MITRE ATT&CK-aligned; rules

3 — Management	CyberShield ODOO Module	Receive events via secure REST API, store with HMAC-SHA256 integrity, provide full SIEM functionality
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2.2 Technology Stack

Component	Technology
ERP Platform	ODOO 18.0 Community Edition
Backend Language	Python 3.10+
Database	PostgreSQL 15
Agent Backend	Wazuh 4.7.x
Windows Agent	PowerShell 5.1+ + Sysmon
Linux Agent	Python 3 + auditd + systemd
Containerisation	Docker + Docker Compose
Reverse Proxy	nginx with TLS 1.3

3. Functional Requirements

3.1 Device Management (FR-DVC)

ID	Requirement	Description	Priority
FR-DVC-01	Device Registration	Shall allow registration of endpoints (workstations, servers, laptops, VMs, network devices)	MUST
FR-DVC-02	UUID Identification	Each device shall have an immutable UUID4 identifier generated at creation	MUST
FR-DVC-03	Agent Token	Each device shall have a unique HMAC-SHA256 agent token for Zero Trust API authentication	MUST
FR-DVC-04	Risk Scoring	System shall calculate a risk score (0-100) per device based on alerts, vulnerabilities and criticality	MUST
FR-DVC-05	Department Grouping	Devices shall be groupable by department (Design, Accounting, Sales, IT, Operations, etc.)	SHOULD

3.2 Audit Logging (FR-LOG)

ID	Requirement	Description	Priority
FR-LOG-01	Event Ingestion	Shall accept security events via REST API from authorised agents	MUST
FR-LOG-02	Immutability	Stored log entries shall be immutable — write and delete operations shall be forbidden at ORM level	MUST
FR-LOG-03	Hash Chaining	Each log entry shall include HMAC-SHA256 hash chained with previous entry hash	MUST
FR-LOG-04	MITRE Mapping	Events shall be classifiable by MITRE ATT&CK; tactic and technique	SHOULD
FR-LOG-05	Integrity Check	System shall provide one-click integrity verification per log entry	SHOULD

3.3 Alert Management (FR-ALT)

ID	Requirement	Description	Priority
FR-ALT-01	Auto-generation	Alerts shall be automatically created for log events with severity HIGH or CRITICAL	MUST
FR-ALT-02	SLA Tracking	CRITICAL=1h, HIGH=4h, MEDIUM=24h, LOW=72h — automatic breach detection	MUST
FR-ALT-03	Lifecycle	New → Acknowledged → Investigating → Contained → Resolved / False Positive	MUST
FR-ALT-04	Auto-Incident	CRITICAL alerts shall automatically create an associated incident	MUST

3.4 Notification Center (FR-NOT)

ID	Requirement	Description	Priority
FR-NOT-01	Multi-channel	ODOO internal, email and webhook notification channels	MUST
FR-NOT-02	No External Deps	Core notifications shall work without any external service dependency	MUST
FR-NOT-03	Escalation Rules	Configurable rules with severity filters, delay, repeat settings and device filters	SHOULD
FR-NOT-04	Full History	All notifications recorded with delivery status, timestamp, error details	MUST
FR-NOT-05	Retry	Failed notifications shall be retryable (max 5 retries)	SHOULD

4. Non-Functional Requirements

4.1 Security (NFR-SEC)

- NFR-SEC-01: All API communications shall use TLS 1.3 minimum in production
- NFR-SEC-02: Every API request shall be authenticated via agent token + HMAC-SHA256 signature
- NFR-SEC-03: Rate limiting enforced at 1,000 events per 60-second window per agent
- NFR-SEC-04: All input shall be validated and sanitised before processing (whitelist approach)
- NFR-SEC-05: Access control shall follow RBAC with 5 differentiated roles
- NFR-SEC-06: Audit logs shall be immutable with cryptographic integrity verification

4.2 Performance (NFR-PER)

- NFR-PER-01: API endpoint shall respond in < 500ms for 95% of requests under normal load
- NFR-PER-02: Dashboard shall load in < 3 seconds with up to 10,000 log entries
- NFR-PER-03: System shall handle up to 100 concurrent agent connections

4.3 Reliability (NFR-REL)

- NFR-REL-01: System shall achieve 99.5% uptime in production deployment
- NFR-REL-02: Failed notifications shall be retried automatically up to 5 times
- NFR-REL-03: Database backups shall be performed daily with 30-day retention